

SIMATS ENGINEERING
CO4 ASSESSMENT TOOL 02

COURSE NAME: Ethical Hacking

COURSE CODE: ITA1404

COURSE OUTCOME COVERED

CO4: Implement network security testing methods by performing web server attacks, analyzing web application vulnerabilities, and assessing wireless network security using Kali Linux.

QUESTIONS: 10

Total Marks:100

Annexure A

VULNERABILITY ASSESSMENT REPORT

Code of Conduct:

I S Venkata Abhishek(192311053) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Abhishek

VULNERABILITY ASSESSMENT REPORT

Wireless Network Security Incident Analysis

Introduction

Wireless networks provide convenient connectivity for employees, customers, guests, and business systems. However, poorly configured wireless infrastructure can expose organizations to unauthorized access, credential theft, session hijacking, malware infection, data leakage, and disruption of services. Vulnerabilities such as weak encryption, open authentication, inadequate network segmentation, rogue access points, insecure captive portals, poor access control, and insufficient monitoring can allow attackers to compromise sensitive systems.

This Vulnerability Assessment Report examines ten major wireless-security incidents and evaluates the security weaknesses associated with each incident. The assessment focuses on wireless encryption, authentication, network segmentation, access control, monitoring, endpoint protection, and protection of sensitive information. Appropriate security controls such as WPA2/WPA3, 802.1X, enterprise authentication, Protected Management Frames, VPNs, multifactor authentication, wireless intrusion detection, network segmentation, and PCI DSS-related controls are recommended to reduce similar risks.

1. TJX Companies Wireless Network Breach, 2005–2007

1.1 Incident Overview

TJX Companies experienced a major payment-card data breach after attackers gained unauthorized access through the company's wireless network. The organization was using WEP during a transition toward stronger wireless security. Inadequate wireless controls, insufficient monitoring, and weak network segmentation contributed to the attackers' ability to reach systems containing sensitive payment information.

1.2 Vulnerability Assessment

Wireless Encryption

The use of WEP represented a major security weakness. WEP uses outdated cryptographic mechanisms and can be compromised with sufficient captured wireless traffic. It should not be used for modern enterprise networks.

Risk: Critical

Authentication

Weak wireless authentication allowed unauthorized devices to obtain network access. Enterprise environments should authenticate individual users and devices instead of relying only on a shared wireless password.

Risk: High

Network Segmentation

The wireless network was not sufficiently isolated from systems containing sensitive payment information. A compromised wireless device could therefore provide a pathway toward internal systems.

Risk: Critical

Monitoring

Insufficient wireless monitoring reduced the organization's ability to detect unauthorized access and suspicious wireless activity.

Risk: High

Payment-Data Protection

Payment-card systems should be isolated from general corporate and wireless networks. Sensitive cardholder information requires strong access controls, encryption, monitoring, and compliance controls.

Risk: Critical

1.3 Recommended Controls

1. Replace WEP with **WPA2-Enterprise or WPA3-Enterprise**.
2. Implement **802.1X authentication** using an appropriate EAP method.
3. Use individual user/device credentials instead of shared passwords.
4. Segment payment-card systems using VLANs and firewalls.
5. Implement wireless intrusion detection and prevention.
6. Deploy rogue-access-point detection.
7. Encrypt sensitive information both in transit and at rest.
8. Apply least-privilege access controls.
9. Maintain centralized security logs and continuous monitoring.
10. Apply relevant **PCI DSS** requirements for cardholder-data environments.
11. Regularly perform vulnerability assessments and penetration testing.
12. Disable unnecessary wireless services and interfaces.

1.4 Risk Rating

Overall Risk: Critical

1.5 Conclusion

The TJX incident demonstrates that wireless security cannot be treated independently from internal network security. Strong encryption alone would not have been sufficient. Enterprise authentication, network segmentation, monitoring, and payment-data security controls should work together to prevent a wireless compromise from becoming a large-scale data breach.

2. Lowe's Wireless Network Intrusion, 2003

2.1 Incident Overview

Attackers compromised a wireless network at a Lowe's retail location in Southfield, Michigan. The wireless connection provided a pathway into broader corporate systems, and attackers attempted to install software intended to capture customers' credit-card information.

2.2 Vulnerability Assessment

Wireless Authentication

The wireless environment did not provide sufficient protection against unauthorized users.

Risk: Critical

Wireless Isolation

The most significant architectural weakness was insufficient separation between the wireless network and sensitive corporate systems.

Risk: Critical

Firewall Controls

Firewalls should have restricted communication from wireless networks to only the systems and services that were actually required.

Risk: High

Point-of-Sale Segmentation

Point-of-sale systems should never be directly reachable from an untrusted wireless network.

Risk: Critical

Intrusion Detection

The absence or inadequacy of intrusion detection allowed suspicious activity to remain undetected.

Risk: High

Centralized Access Management

Wireless access should have been centrally controlled and monitored so that unauthorized devices could be identified and removed.

Risk: High

2.3 Recommended Security Architecture

A secure retail architecture should follow this structure:

Wireless Client → Access Point → Wireless VLAN → Firewall → Restricted Application Services

The POS environment should be placed in a separate protected segment:

POS Network → Internal Firewall → Payment Processing Systems

Direct communication between guest/wireless networks and POS systems should be prohibited.

2.4 Recommended Controls

- WPA2-Enterprise/WPA3-Enterprise
- 802.1X authentication
- Network access control
- VLAN segmentation
- Internal firewalls
- IDS/IPS
- Centralized authentication
- Centralized logging
- Endpoint security
- Regular vulnerability assessments
- Strict POS access policies

2.5 Risk Rating

Overall Risk: Critical

2.6 Conclusion

The Lowe's incident demonstrates that a local wireless compromise can become a corporate-wide security problem when wireless networks are trusted too much. Strong authentication combined with segmentation and firewall restrictions would significantly reduce the possibility of lateral movement.

3. Firesheep and Public Wi-Fi Session Hijacking, 2010

3.1 Incident Overview

Firesheep demonstrated the dangers of using web applications over open wireless networks when session information was transmitted without adequate encryption. An attacker connected to the same wireless network could potentially capture unprotected session cookies and impersonate authenticated users.

3.2 Vulnerability Assessment

Open Wi-Fi Authentication

Open Wi-Fi networks provide little protection against other users on the same network.

Risk: High

Session-Cookie Protection

Session cookies transmitted without encryption could potentially be captured and reused.

Risk: Critical

HTTPS Enforcement

Websites that protected only the login process but later transmitted session information over unencrypted HTTP remained vulnerable.

Risk: Critical

HSTS

HTTP Strict Transport Security helps browsers consistently use HTTPS and reduces the possibility of protocol downgrade attacks.

Risk: High

VPN

A VPN can provide an additional encrypted tunnel when users are connected to an untrusted network.

Risk Reduction: High

Wireless Client Isolation

Client isolation prevents wireless clients from directly communicating with one another and reduces local network attacks.

Risk Reduction: High

3.3 Recommended Controls

1. Enforce HTTPS for the entire web session.
2. Enable HSTS.
3. Use secure and HttpOnly cookie attributes.
4. Rotate session identifiers after authentication.
5. Implement appropriate session expiration.
6. Use TLS 1.2/1.3 with properly validated certificates.
7. Enable wireless client isolation.

8. Recommend VPN usage on untrusted networks.
9. Educate users about public Wi-Fi risks.
10. Avoid transmitting sensitive information over unencrypted connections.

3.4 Risk Rating

Overall Risk: High

3.5 Conclusion

The Firesheep incident demonstrated that protecting only the login page is insufficient. The complete authenticated session must remain encrypted. HTTPS, HSTS, secure cookies, client isolation, VPN usage, and user awareness together provide significantly stronger protection.

4. DarkHotel Attacks Through Hotel Wi-Fi Networks

4.1 Incident Overview

The DarkHotel campaign demonstrated the risks faced by business executives and other high-value individuals using hotel networks. Attackers could exploit compromised hotel infrastructure and fraudulent software-update mechanisms to deliver malicious software to selected victims.

4.2 Vulnerability Assessment

Guest Authentication

Hotel networks that rely on weak or poorly designed authentication may allow unauthorized users to access the network.

Risk: High

Captive Portal Security

A captive portal must be securely designed and separated from internal hotel systems. Administrative interfaces should never be exposed through the guest network.

Risk: High

DNS Integrity

Compromised DNS services could redirect users toward malicious destinations.

Risk: Critical

Network Segmentation

Guest devices must be isolated from hotel management systems, staff systems, cameras, payment systems, and administrative infrastructure.

Risk: Critical

Third-Party Access

Hotel networks frequently involve external service providers. Poorly controlled third-party accounts can create additional attack paths.

Risk: High

Software-Update Verification

Users and endpoints should verify digital signatures and certificates before accepting software updates.

Risk: Critical

Malware Detection

Endpoint detection and response, antivirus, application control, and network monitoring should be implemented.

Risk: High

4.3 Recommended Controls

- WPA2/WPA3 security for appropriate wireless networks
- Guest-network isolation
- Separate VLANs for guests, employees, IoT devices, and management
- Secure DNS and DNS monitoring
- MFA for administrators
- Strict third-party access controls
- Digitally signed software updates
- Endpoint detection and response
- Wireless intrusion detection
- Centralized security monitoring
- Regular security assessments

4.4 Risk Rating

Overall Risk: Critical

4.5 Conclusion

Hotel Wi-Fi is a high-risk environment because it contains large numbers of unknown devices and may involve multiple third-party service providers. Strong segmentation, secure update mechanisms, DNS protection, endpoint security, and continuous monitoring are essential.

5. Marriott Wi-Fi Deauthentication Incident, 2014

5.1 Incident Overview

The Marriott incident involved the use of Wi-Fi deauthentication messages that disrupted guests' personal hotspots. The incident demonstrated that IEEE 802.11 management frames can create availability and connectivity risks when appropriate protections are not implemented.

5.2 Vulnerability Assessment

Deauthentication Attacks

Traditional Wi-Fi management frames were not adequately protected against spoofing. An attacker could potentially send forged deauthentication frames to disconnect wireless clients.

Risk: High

Availability

Repeated disconnection can prevent legitimate users from accessing wireless services.

Risk: High

Wireless Interference

Unauthorized wireless activity can affect network availability and reliability.

Risk: High

Protected Management Frames

Protected Management Frames (PMF), defined by IEEE 802.11 standards, provide protection for important management traffic.

Risk Reduction: High

Wireless Intrusion Detection

A WIDS/WIPS can detect abnormal deauthentication activity and other wireless attacks.

Risk Reduction: High

5.3 Recommended Controls

1. Deploy WPA3 where supported.
2. Enable Protected Management Frames.
3. Use appropriate WPA2 configurations where WPA3 is unavailable.
4. Deploy wireless intrusion detection/prevention.
5. Monitor unusual management-frame activity.

6. Establish acceptable-use policies.
7. Maintain incident-response procedures.
8. Restrict unauthorized wireless equipment.
9. Perform regular wireless security assessments.
10. Ensure security controls comply with applicable laws and regulations.

5.4 Legal and Compliance Considerations

Wireless interference and intentional disruption can create legal and regulatory consequences. Organizations should ensure that wireless security testing and monitoring are performed only within authorized environments.

5.5 Risk Rating

Overall Risk: High

5.6 Conclusion

The incident highlights that wireless confidentiality is not the only concern; availability is also important. PMF, WPA3, WIDS/WIPS, monitoring, and clear acceptable-use policies can reduce deauthentication and wireless-disruption risks.

6. Fake Wi-Fi Networks at the Republican National Convention, 2016

6.1 Incident Overview

A security experiment during the 2016 Republican National Convention established fake open wireless networks near the event venue. A large number of users connected to the networks, demonstrating how easily users can trust familiar-looking SSIDs.

6.2 Vulnerability Assessment

SSID Trust

Users may connect to a network simply because its SSID appears legitimate.

Risk: High

Automatic Reconnection

Devices that automatically reconnect to previously known networks can connect to malicious access points.

Risk: High

Captive-Portal Authentication

Fake portals can be designed to collect credentials or personal information.

Risk: Critical

Evil-Twin Attacks

An attacker can imitate the SSID of a legitimate wireless network and attract users.

Risk: Critical

Certificate Validation

Users may ignore certificate warnings or fail to verify whether a connection is legitimate.

Risk: High

VPN Enforcement

VPN usage can protect traffic from local network observers when appropriately configured.

Risk Reduction: High

6.3 Recommended Controls

- Disable automatic connection to unknown networks.
- Use WPA2/WPA3-Enterprise where possible.
- Implement wireless monitoring.
- Detect duplicate or rogue SSIDs.
- Use certificate-based authentication.
- Educate users about fake access points.
- Use VPNs on untrusted networks.
- Do not ignore certificate warnings.
- Avoid entering sensitive information into suspicious captive portals.
- Establish incident-reporting procedures.

6.4 Risk Rating

Overall Risk: High

6.5 Conclusion

The incident shows that users are an important part of wireless security. Even technically strong infrastructure can be bypassed if users automatically connect to malicious networks. Technical controls must therefore be combined with awareness and device-management policies.

7. Wi-Fi Attacks Against Anti-Doping Officials, 2016

7.1 Incident Overview

Close-access teams reportedly targeted Wi-Fi networks used by anti-doping officials during international sporting events. Hotel and event wireless networks were used as part of attempts to obtain credentials and access sensitive information.

7.2 Vulnerability Assessment

Public-Network Authentication

Public Wi-Fi should not be treated as a trusted corporate environment.

Risk: High

Credential Protection

Passwords entered over compromised networks may be exposed through phishing, malware, or other attacks.

Risk: Critical

Multifactor Authentication

MFA reduces the consequences of password compromise.

Risk Reduction: Critical

Device Certificates

Managed devices can use certificates to authenticate to enterprise wireless networks.

Risk Reduction: High

Endpoint Security

Devices containing sensitive information require EDR, anti-malware, encryption, and patch management.

Risk: High

VPN

A properly configured VPN provides encrypted communication over untrusted networks.

Risk Reduction: High

Travel Security

Personnel traveling internationally should receive security guidance before using hotel, airport, or event Wi-Fi.

Risk: High

Medical Information

Medical and anti-doping information is highly sensitive and should be protected through strict access controls and encryption.

Risk: Critical

7.3 Recommended Controls

1. Enforce MFA for important systems.
2. Use device certificates for managed devices.
3. Require VPN connections outside trusted networks.
4. Encrypt sensitive information.
5. Deploy EDR.
6. Implement mobile-device management.
7. Apply strong password and credential policies.
8. Provide travel-security training.
9. Restrict access based on least privilege.
10. Monitor authentication logs for suspicious activity.
11. Use remote-wipe capabilities for managed devices.
12. Separate sensitive systems from public Wi-Fi.

7.4 Risk Rating

Overall Risk: Critical

7.5 Conclusion

Personnel working with sensitive information should assume that hotel and event Wi-Fi is untrusted. MFA, device certificates, VPNs, endpoint security, and travel-security policies significantly reduce the impact of wireless attacks.

8. Attempted OPCW Wi-Fi Intrusion, 2018

8.1 Incident Overview

In April 2018, Dutch authorities disrupted an attempted close-access cyber operation targeting the Organisation for the Prohibition of Chemical Weapons in The Hague. Equipment intended for a Wi-Fi compromise was reportedly found in a vehicle near the organization's premises.

8.2 Vulnerability Assessment

Radio-Frequency Exposure

Wireless signals can extend beyond the physical boundaries of an organization. Attackers located outside the building may therefore attempt to interact with wireless infrastructure.

Risk: High

Enterprise Wi-Fi Authentication

Shared passwords provide weaker accountability than enterprise authentication.

Risk: High

802.1X

802.1X provides port-based network access control and can authenticate individual users or devices.

Risk Reduction: Critical

EAP Certificate Validation

Improper certificate validation can allow clients to connect to fraudulent authentication infrastructure.

Risk: Critical

Rogue-Device Detection

Unauthorized access points and wireless devices should be detected quickly.

Risk: High

Wireless Intrusion Prevention

WIPS can identify suspicious wireless behavior and support rapid response.

Risk Reduction: High

Physical Surveillance

Physical security cameras and monitoring can identify equipment positioned near facilities for wireless attacks.

Risk Reduction: High

Network Access Control

NAC can verify device identity and security posture before allowing network access.

Risk Reduction: High

8.3 Recommended Controls

- WPA3-Enterprise
- 802.1X
- Certificate-based EAP authentication
- Proper certificate validation
- Network Access Control
- WIDS/WIPS
- Rogue AP detection

- RF monitoring
- Directional antenna and signal-management considerations
- Physical surveillance
- Security patrols around sensitive areas
- Centralized wireless logging

8.4 Risk Rating

Overall Risk: Critical

8.5 Conclusion

Organizations handling sensitive government or scientific information must protect both the logical and physical aspects of wireless security. Strong enterprise authentication should be combined with RF monitoring, rogue-device detection, physical surveillance, and network-access control.

9. Australian Airport and In-Flight Evil-Twin Case, 2024

9.1 Incident Overview

Australian authorities charged an individual who allegedly created fake Wi-Fi access points that imitated legitimate networks at airports and other locations. The incident demonstrated the risks associated with SSID impersonation and users automatically connecting to fraudulent wireless networks.

9.2 Vulnerability Assessment

SSID Impersonation

SSID names alone cannot prove that a wireless network is legitimate.

Risk: High

Automatic Connection

Devices configured to automatically join known networks can connect to an attacker's access point.

Risk: High

Captive Portal

A fake captive portal can attempt to collect names, email addresses, passwords, or other personal information.

Risk: Critical

Certificate Warnings

Ignoring browser certificate warnings can expose users to phishing and interception.

Risk: High

Passenger-Data Exposure

Airports contain large numbers of users transmitting personal and business information.

Risk: High

Rogue-Access-Point Detection

Airports and airlines should continuously monitor the RF environment for unauthorized access points.

Risk: High

9.3 Recommended Controls

1. Deploy WIDS/WIPS.
2. Monitor for duplicate SSIDs.
3. Maintain an official list of legitimate wireless networks.
4. Educate passengers and staff.
5. Disable automatic connection to unknown networks.
6. Use WPA3-Enterprise for staff networks.
7. Use secure captive portals with HTTPS.
8. Never instruct users to bypass certificate warnings.
9. Require VPN for sensitive staff communications.
10. Establish procedures for reporting suspicious wireless networks.
11. Train airport personnel to respond to rogue AP incidents.

9.4 Risk Rating

Overall Risk: High

9.5 Conclusion

The airport environment is attractive to attackers because thousands of users connect to wireless networks every day. Evil-twin detection, secure authentication, user awareness, and rapid staff response are essential to protect passengers and employees.

10. Network Rail Public Wi-Fi Portal Compromise, 2024

10.1 Incident Overview

On September 25, 2024, public Wi-Fi services at 19 major UK railway stations displayed unauthorized offensive content. Reports indicated that the Wi-Fi landing page had been changed using a legitimate administrator account. The incident demonstrates the danger of compromised privileged accounts and weak administrative access controls.

10.2 Vulnerability Assessment

Privileged Access Management

Administrative accounts have extensive privileges and therefore require stronger security controls.

Risk: Critical

Multifactor Authentication

A single password is not sufficient protection for highly privileged accounts.

Risk: Critical

Captive-Portal Security

The public-facing portal should be isolated from the administrative management environment.

Risk: High

Third-Party Supplier Access

Public Wi-Fi services are often operated with support from external suppliers. Excessive or poorly controlled supplier privileges increase risk.

Risk: High

Logging

Administrative activities should be centrally logged and monitored.

Risk: High

Change Approval

Changes to public-facing content should follow an approved change-management process.

Risk: High

Account Monitoring

Privileged accounts should be monitored for unusual login locations, times, devices, and activities.

Risk: Critical

Incident Response

Organizations must be capable of disabling compromised accounts, isolating affected systems, restoring trusted configurations, and preserving evidence.

Risk: High

10.3 Recommended Controls

1. Implement **Privileged Access Management (PAM)**.
2. Enforce MFA for every administrator.
3. Use separate administrative accounts rather than normal user accounts.
4. Apply least privilege.
5. Review third-party accounts regularly.
6. Disable unused supplier accounts immediately.
7. Use just-in-time privileged access where practical.
8. Maintain immutable or protected audit logs.
9. Monitor privileged logins through a SIEM.
10. Require change approval for public-facing portal modifications.
11. Implement configuration integrity monitoring.
12. Maintain tested backup and restoration procedures.
13. Establish a formal incident-response plan.
14. Conduct periodic access reviews.

10.4 Risk Rating

Overall Risk: Critical

10.5 Conclusion

The Network Rail incident demonstrates that legitimate accounts can become attack paths when credentials are compromised or access controls are inadequate. Strong MFA, PAM, least privilege, supplier management, logging, change control, and rapid incident response can substantially reduce this risk.

Overall Comparative Risk Assessment

Case	Primary Vulnerability	Main Impact	Risk
TJX	WEP and poor segmentation	Payment-card data breach	Critical
Lowe's	Wireless-to-corporate access	Lateral movement/POS compromise	Critical

Firesheep	Unencrypted sessions	Session hijacking	High
DarkHotel	Compromised hotel Wi-Fi	Malware/credential theft	Critical
Marriott	Deauthentication frames	Availability disruption	High
Fake Wi-Fi	Evil-twin networks	Credential/data exposure	High
Anti-Doping Wi-Fi	Credential compromise	Sensitive-data theft	Critical
OPCW	Close-access Wi-Fi attack	Unauthorized network access	Critical
Australian Airport	Fake access points	Personal-data exposure	High
Network Rail	Privileged account compromise	Portal manipulation	Critical

Common Vulnerabilities Identified

The ten incidents demonstrate several recurring wireless-security weaknesses:

1. Weak Encryption

Legacy technologies such as WEP are unsuitable for modern enterprise networks. Organizations should migrate to WPA2-Enterprise or WPA3-Enterprise as appropriate.

2. Weak Authentication

Shared passwords and password-only authentication provide limited accountability. Enterprise wireless environments should use 802.1X and centralized authentication.

3. Poor Network Segmentation

Wireless users should not automatically receive access to corporate, payment, administrative, or operational systems.

4. Insufficient Monitoring

Organizations need WIDS/WIPS, centralized logging, SIEM monitoring, and alerting to detect suspicious wireless activity.

5. Rogue Access Points

Unauthorized access points can create an alternative path into an organization's network or imitate legitimate wireless services.

6. Inadequate Endpoint Protection

Devices connecting to public wireless networks should have current operating systems, security software, encryption, and appropriate endpoint-management controls.

7. Weak Privileged Access Management

Administrative accounts require MFA, least privilege, monitoring, access reviews, and controlled changes.

8. Lack of User Awareness

Users may connect to fake SSIDs, ignore certificate warnings, reuse passwords, or enter credentials into fraudulent captive portals.

Recommended Enterprise Wireless Security Architecture

A secure enterprise wireless environment should follow a layered security model:

User Device

↓

WPA3/WPA2-Enterprise

↓

802.1X + RADIUS Authentication

↓

Wireless Access Point

↓

VLAN / Network Access Control

↓

Firewall

↓

Restricted Internal Services

Sensitive environments should be further separated:

Guest Network → Internet Only

Employee Network → Authorized Corporate Services

IoT Network → Restricted IoT Services

POS Network → Payment Processing Systems

Management Network → Authorized Administrators Only

This architecture prevents a compromise in one wireless segment from automatically providing access to sensitive systems.

Priority Security Recommendations

Immediate Priority

- Replace WEP and other obsolete wireless security.
- Enable WPA2-Enterprise/WPA3-Enterprise.
- Implement 802.1X.

- Enable MFA for administrative and sensitive accounts.
- Separate guest, employee, POS, IoT, and management networks.
- Deploy rogue-access-point detection.
- Enable wireless intrusion detection/prevention.
- Enforce HTTPS and secure session management.
- Implement VPN for sensitive communication over untrusted networks.
- Protect privileged accounts using PAM.

Medium-Term Improvements

- Implement centralized SIEM monitoring.
- Introduce network access control.
- Deploy endpoint detection and response.
- Conduct regular wireless vulnerability assessments.
- Perform access-control reviews.
- Improve third-party/vendor access management.
- Develop wireless incident-response procedures.
- Conduct security-awareness training.

Long-Term Improvements

- Adopt a Zero Trust approach.
- Continuously monitor wireless infrastructure.
- Implement certificate-based device authentication.
- Conduct regular penetration testing within authorized scope.
- Maintain security baselines and configuration standards.
- Perform regular security audits and compliance assessments.
- Continuously review emerging wireless threats.

Conclusion

The ten wireless-security incidents demonstrate that wireless vulnerabilities can affect confidentiality, integrity, and availability. The TJX and Lowe's incidents show how weak wireless security and poor network segmentation can expose payment and corporate systems. The Firesheep incident demonstrates the importance of complete HTTPS protection and secure session management. DarkHotel highlights the risks of compromised third-party Wi-Fi environments and malicious software-

update mechanisms. The Marriott incident demonstrates that wireless management frames can create availability problems when they are not adequately protected.

The fake Wi-Fi and Australian airport incidents demonstrate the effectiveness of evil-twin attacks against users who trust familiar SSIDs. The anti-doping and OPCW cases demonstrate that attackers can perform close-access wireless operations against organizations handling highly sensitive information. Finally, the Network Rail incident shows that wireless security also depends on strong privileged-access management, MFA, monitoring, supplier control, and change management.

Therefore, a comprehensive wireless security strategy should not rely on encryption alone. **WPA2/WPA3, 802.1X, MFA, network segmentation, Protected Management Frames, VPNs, rogue-access-point detection, WIDS/WIPS, endpoint protection, privileged-access management, continuous monitoring, user awareness, and appropriate compliance controls** should operate together.

A properly designed defense-in-depth architecture can significantly reduce the probability that a wireless compromise will develop into a major organizational security incident.